

## Requirements:

1. Create a database of significant market events from the past 3 years including:
  - Earnings announcements (at least 10 instances)
  - Federal Reserve interest rate decisions (all instances)
  - Major economic report releases (GDP, employment, inflation)
  - Store each event with its date, type, and any relevant numeric data (e.g., actual vs. expected figures)
2. Develop a data collection function that:
  - Fetches price data for multiple assets (at least 5 stocks, 1 index, 1 cryptocurrency)
  - Gets data spanning from 5 trading days before to 5 trading days after each event
  - Ensures data quality through appropriate cleaning techniques
3. Build an event impact analysis system that:
  - Calculates price and volatility changes before, during, and after events
  - Compares actual market reactions to consensus expectations
  - Identifies which assets are most sensitive to which types of events
  - Measures how long the impact of different events typically lasts
4. Implement visualization capabilities:
  - Create event study charts showing average price movement patterns
  - Display volatility changes around events